

Section A: Structured Questions

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Question 1 [10 marks]

(a) Figure 1.1 shows part of the cell body of a neurone.

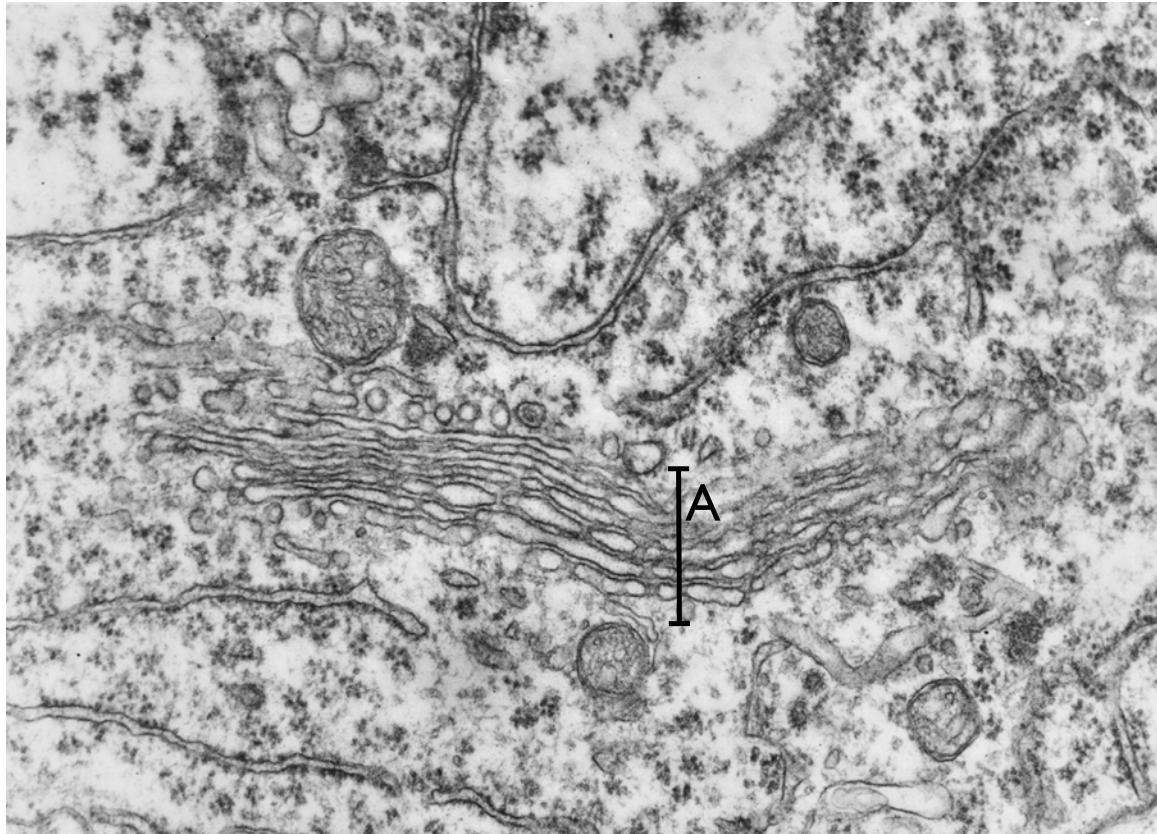


Figure 1.1

(i) Identify the organelle labeled A. [1]

Golgi Apparatus

(ii) Relate the importance of this organelle to the cell. [2]

- a. **Post-translational modification by chemical/physical modification**
- b. **e.g. modifying proteins that would form the ion channels in the membrane of the neurone**
- c. **synthesizing and packaging neurotransmitters to be sent to the synaptic knob**
- d. **ion channels/transmembrane transport proteins are embedded in membrane of golgi vesicle which also replenishes the membrane at the synaptic knob**

The following structure in Figure 1.2 is also found in the axon of a neurone.

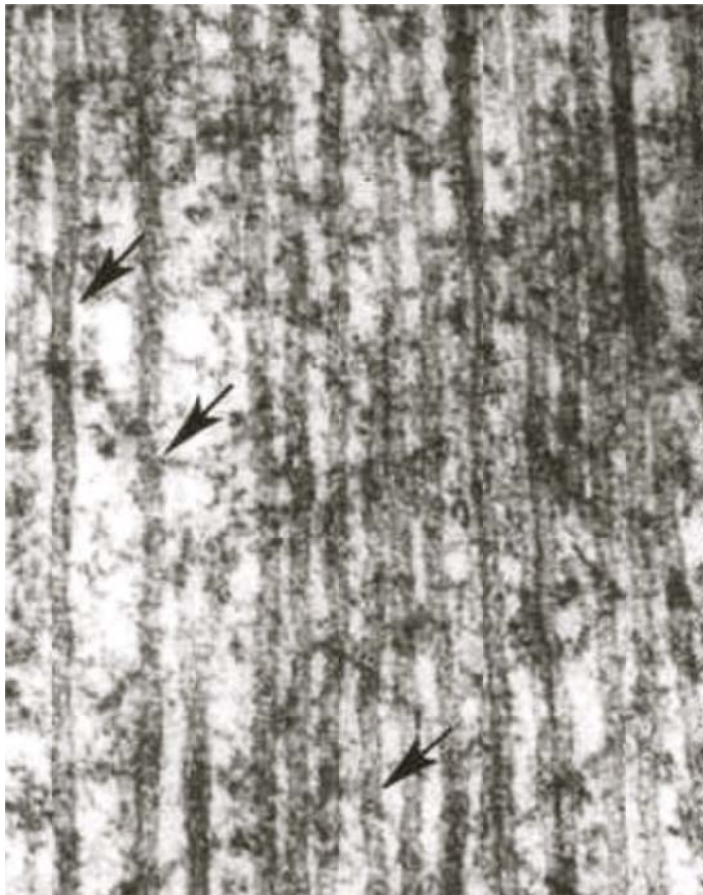


Figure 1.2

(iii) Identify the structures as indicated by the arrows in the diagram. [1]

Microtubules

(iv) In what way is this structure related to the function of organelle A in the neurone? [1]

- a. to transport golgi vesicles to the respective parts of the neurone**
- b. with the use of ATP OR**
- c. repair and maintenance of the membrane**

(b) Figure 1.3 shows a cholinergic synapse of the neurone.

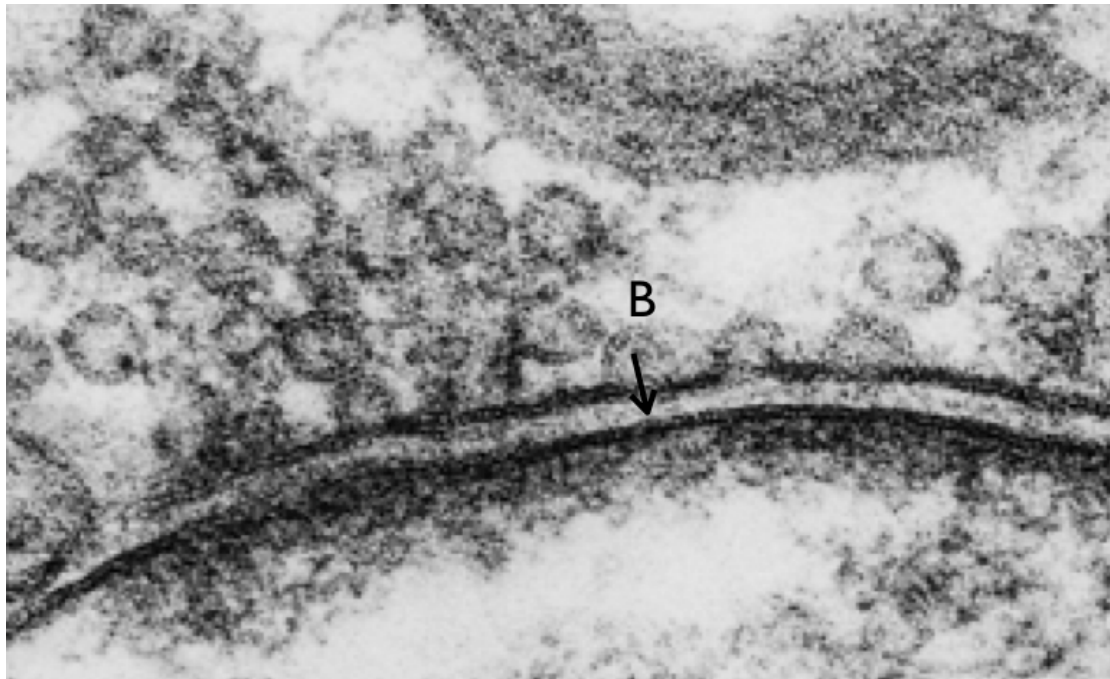


Figure 1.3

(i) Identify process B. [1]

Exocytosis

(ii) With your understanding of the cell membrane, explain how process B occurs. [1]

- a. **Phospholipid bilayer is fluid due to weak hydrophobic interactions between hydrophobic hydrocarbon tails,**
- b. **allowing synaptic vesicles can thus fuse with the pre-synaptic membrane**

(iii) Suggest the significance of this process with regard to synaptic transmission. [1]

- a. **Neurotransmitters must be released into the synaptic cleft in bulk to transmit information from the impulse quickly to the next neurone;;**
OWTTE.

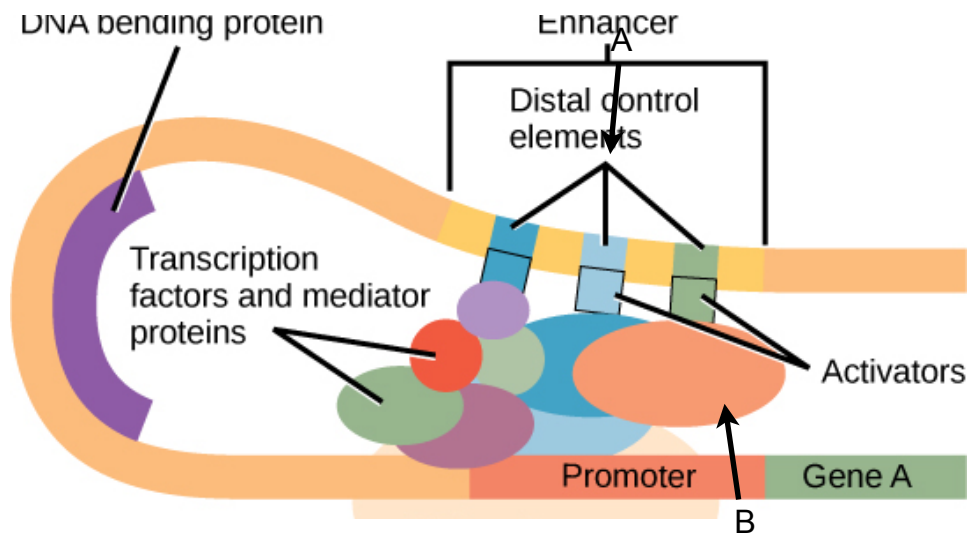
(iv) On the pre-synaptic membrane, there are also Ca^{2+} voltage-gated channels found. Describe how a change in potential difference will cause them to open. [2]

- a. **When the potential difference across the membrane is varied from the resting potential,**
- b. **conformational change in the protein causes the gate to open**
- c. **This allows charged ions to flow down their electrochemical gradient**
- d. **through the hydrophilic channel provided by this transmembrane protein**

Question 2 [13 marks]

(a) Figure 2.1 below shows the transcription initiation process.

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Figure 2.1

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(i) Identify structures A and B [1]

A: Enhancer

B: Promoter

(ii) With reference to Figure 2.1, describe how transcription is initiated and regulated. [3]

- Binding of transcription factor to TATA box
- Recruiting of other TFs and RNA polymerase to bind at the promoter to form transcription initiation complex
- Transcription can proceed at a basal rate
- Activator binds to the enhancer
- DNA bends to allow activator protein to bring it closer to the TIC with the help of the DNA bending protein
- Mediator proteins help in ensuring activator binds to TIC
- RNA polymerase is stabilized/increase affinity to promoter and it is able to transcribe at a higher rate.

{max 3}

- (b) Figure 2.2 is an electron micrograph depicting the process of translation in a prokaryote.

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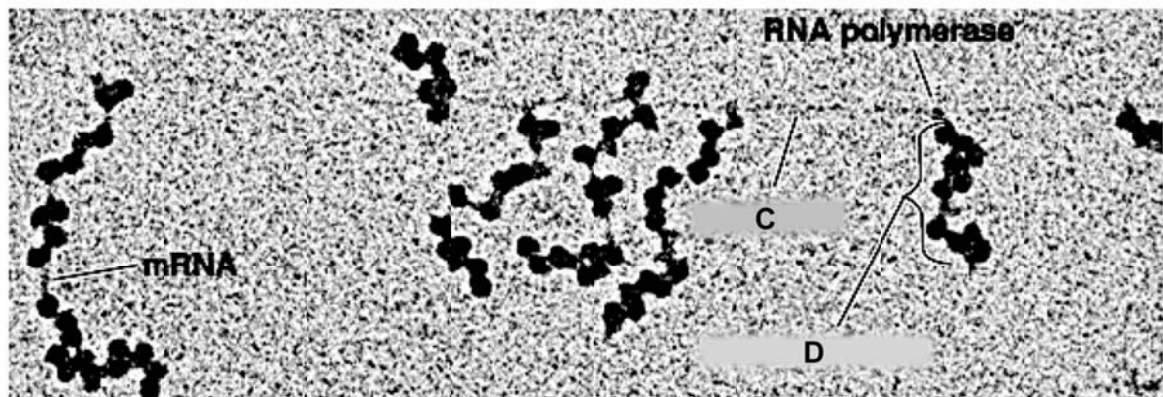


Figure 2.2

- (i) Identify structures C and D. [1]

C: DNA

D: Polyribosomes

- (ii) Describe the importance of structure D in translation. [1]

- It increases the rate of translation as many ribosomes are translating the mRNA at the same time;**
- Increase the rate of protein production;**

- (iii) With reference to Figure 2.2, suggest a difference between the translation in prokaryotes and eukaryotes. [1]

- Translation in prokaryotes occur simultaneously with transcription while eukaryotic transcription is in the nucleus and translation occurs separately in the ribosomes in the cytoplasm.**

- (c) Different genetic mutations on a gene and its mRNA may result in different diseases. For example, RNA splice site mutation of β -globin mRNA is responsible of some cases of β -thalassemia and a point mutation on β -globin gene is responsible for sickle cell anaemia.

- (i) Explain the significance of the **point** mutation to the properties of haemoglobin. [3]

- Base A was changed to base T**
- Hydrophilic glutamic acid was replaced with hydrophobic valine;**
- Changing the tertiary structure of haemoglobin as different bonds are formed between the R-groups;**
- low oxygen concentrations, this substitution creates a hydrophobic spot on the outside of the protein structure;**
- that sticks to the hydrophobic region of an adjacent haemoglobin molecule's beta chain**
- This clumping together (polymerization) of Hb S molecules;**
- forms rigid fibers resulting in crystals that distorts the cell;**
- unable to transport oxygen efficiently;**

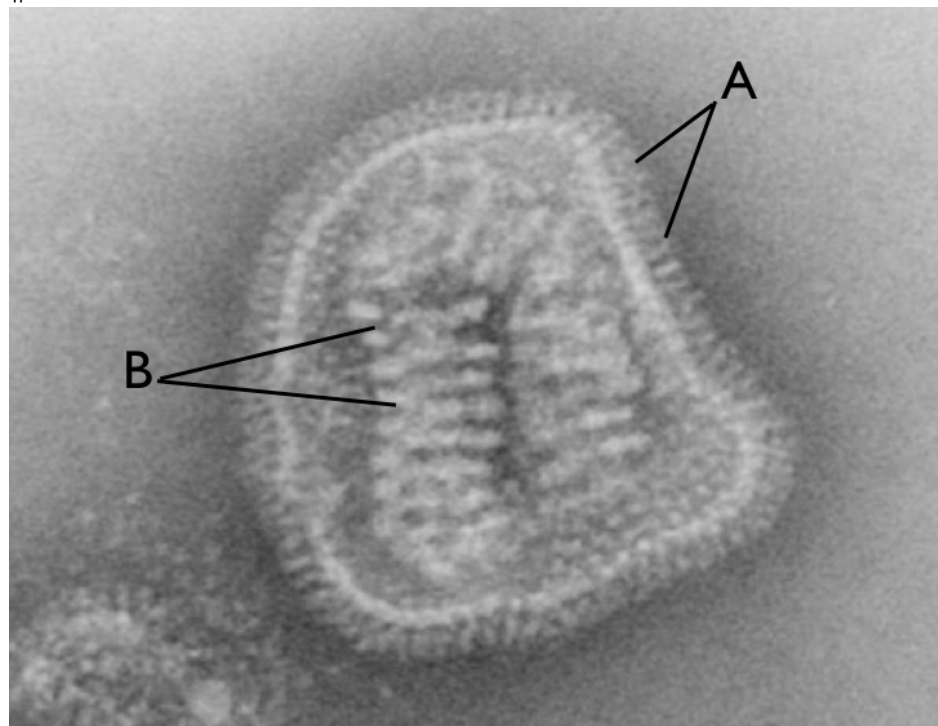
(ii) Explain why a point mutation on a β -globin gene does not always produce mutated globin whereas an RNA splice site mutation will always produce mutated globin. [3]

- a. Point mutation may result in a silent mutation
- b. Where a substitution mutation codes for the same amino acid;
- c. Due to the degeneracy of the triplet code;
- d. An RNA splice site mutation will not allow the spliceosome to recognize and bind to the splice site;
- e. Therefore an intron is not excised leading to a longer mRNA sequence
- f. The introns are translated and leading to a different amino acid sequence (resulting in a mutated globin chain).

Question 3 [13 marks]

Figure 3.1 shows an electron micrograph of H1N1.

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Figure 3.1

(a) Identify the components labeled A and B. [3]

A

- a. Neuraminidase;;
- b. Haemagglutinin;;

B

Negative/antisense RNA segments;;

(b) RNA-dependent RNA polymerase can be found in this virion while reverse transcriptase can be found in retrovirus. Besides the type of virus, describe one similarity and two differences between RNA-dependent RNA polymerase and reverse transcriptase. [3]

Similarity:

- a. Both are viral proteins, i.e. they are only coded for by viral genes and not the host genes.
- b. Both are dependent on RNA as a template

Differences

Points	RdRp	Reverse Transcriptase
a. Template used	RNA (both antisense and sense)	RNA template, then using the single-stranded cDNA
b. Products	Single stranded RNA	Double stranded cDNA
c. Catalytic activity	RNA polymerase – extension of RNA using host free ribonucleotides	Reverse transcription – extension of cDNA Ribonuclease/RNase – degrade RNA in cDNA-RNA DNA polymerase – formation of complementary DNA to form double-stranded cDNA
d. Monomers	Ribonucleotides	Deoxyribonucleotides

Sialic acid

Virion

Oseltamivir

The image displays two chemical structures side-by-side. On the left is Sialic acid, a nine-carbon sugar with a carboxylic acid group at C1, hydroxyl groups at C2, C3, and C6, an amide group at C4, and a methyl group at C5. On the right is Oseltamivir, a synthetic neuraminidase inhibitor. It features a cyclohexene ring with a carboxylic acid group at C1, a hydroxyl group at C2, an amide group at C3, and a methyl group at C4. The C5 position is substituted with an ethoxy group and a side chain consisting of an isopropyl group and a methyl group.

Sialic acid

Oseltamivir

(c) Explain how oseltamivir affects the reproduction cycle of H1N1. [2]

- Oseltamivir has similar conformation and chemical compatibility as sialic acid
- It is thus able to bind to the active site of neuraminidase
- Prevents neuraminidase from cleaving sialic acid
- Hence virion will not be able to be released via budding to infect other cells, (preventing the virus from spreading to the rest of the respiratory tract)

The effects of oseltamivir were studied on patients infected with influenza virus. These patients were randomly separated into two groups – one group administered with a placebo (a pseudo-drug with no effect) while the other group was administered oseltamivir. Symptoms were measured and recorded in Table 3.4.

Symptoms	Study Groups	
	Placebo (n=129)	Oseltamivir, 75mg (n=124)
Cough		
Duration, hr	55 (36-73)	31 (24-42)
Severity, arbitrary units	110	67
Nasal Obstruction		
Duration, hr	43 (31-64)	33 (25-43)
Severity, arbitrary units	80	60
Sore Throat		
Duration, hr	21 (8-29)	10 (4-19)
Severity, arbitrary units	29	20
Fatigue		
Duration, hr	41 (26-50)	24 (19-34)
Severity, arbitrary units	89	54
Feverishness		
Duration, hr	23 (16-29)	10 (8-14)
Severity, arbitrary units	49	21

Table 3.4

Table abstracted from: <http://jama.jamanetwork.com/article.aspx?articleid=192425>,
Efficacy and Safety of the Oral Neuraminidase Inhibitor Oseltamivir in Treating Acute Influenza - A Randomized Controlled Trial

(d) With reference Table 3.4, determine the effectiveness of oseltamivir. [2]

- Oseltamivir is highly effective in treating influenza.**
- Duration of symptoms was reduced in the group administered with oseltamivir. Quote an example from the table. (For e.g. duration of cough was reduced from an average of 52 to 31 hours)**
- Severity was significantly reduced in the group administered with oseltamivir. Quote an example from the table. (For e.g. severity of cough was reduced from 110 to 67 arbitrary units.)**
- The reduction in symptoms indicates that the spread of influenza virus is controlled with the drug, allowing the immune system of the body to target infected cells before the virus can infect even more cells.**

(e) In a separate case, a patient was treated for both H1N1 and H3N3. During this treatment, he was infected with H3N3. Upon testing, it was found that his blood now contains H3N3 and a new strain of virus, H1N3. Explain how this new strain could have arisen. [3]

- The new strain of virus could have arisen due to antigenic shift.**
- As H1N1 and H3N3 infect the same cell simultaneously, their protein capsids and lipid envelopes are removed, exposing their RNA.**
- Influenza virus can undergo antigenic shift because it contains a segmented genome composed of 8 different segments.**
- The new strain, H1N3, could have arisen via genetic recombination of the genome segments.**

- e. The random assembly of RNA segments from two different viruses
 f. resulting in a mixture of surface antigens of both H1N1 and H3N3 strains.

Question 4 [10 marks]

- (a) In maize plants, a recessive mutant allele leads to green leaves as opposed to bronze leaves. Another recessive mutant allele at a separate locus leads to yellow cobs as opposed to brown cobs. F1 maize plants from a standard dihybrid cross was test crossed with a plant with green leaves and yellow cobs. The table shows the phenotypes of 200 of the offspring.

Bronze leaves and yellow cobs	88
Bronze leaves and brown cobs	12
Green leaves and yellow cobs	14
Green leaves and brown cobs	86

- (i) In the space below, draw a genetic diagram to explain the observed results of the cross. [4]

Let **B** be the dominant allele coding for bronze leaves and let **b** be the recessive allele for green leaves.

Let **N** be the dominant allele for brown cobs and let **n** be the recessive allele for yellow cobs. **a. legend [1/2]**

Parental phenotype Bronze leaves, brown cobs x green leaves, yellow cobs **b. correct phenotypes [1/2]**

Genotype $\frac{Bn}{bN}$ x $\frac{bn}{bn}$ **c. correct genotype [1/2]**

Types of Gametes $\frac{Bn}{BN}$ $\frac{bN}{bn}$ bn

d. gametes are circled with arrows [1/2]

e. correct gametic genotype [1/2]

Offspring genotypes & phenotypes **d. [1 mark]**

	Bn	bN	BN	bn
bn (a)	$\frac{Bn}{bn}$ Bronze leaves, yellow cobs	$\frac{bN}{bn}$ Green leaves, brown cobs	$\frac{BN}{bn}$ Bronze leaves, brown cobs	$\frac{bn}{bn}$ Green leaves, yellow cobs

Expected ratio: 1: 1: 1: 1
 Experimental ratio 44% 43% 6% 7%

(parental) (parental) (recombinant) (recombinant)

f. experimental ratio [1/2]

- (ii) Explain the results of the test cross. [3]
- Bulk of experimental plants has parental phenotypes of bronze leaves + yellow cobs and green leaves and brown cobs for test cross.
 - This means that the two genes (for leaf colour and cob colour) are linked and found on the same chromosomes.
 - Higher probability that parental phenotypes will predominate. This is because linked genes do not assort independently.
 - The bronze leaves and brown cobs and green leaves and purple cobs phenotypes are recombinant phenotypes as a result of crossing over.
 - Since crossing over is a chance event,
 - the recombination phenotype (due to crossing over) to be lower in frequency than parental phenotype.

- (b) A chi-squared test was also carried out for the above cross and the chi-square value was 109.6. Using the probability table given, does this value for the chi-squared test support the expected phenotypic ratio? Explain. [3]

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

- No;
- Calculated χ^2 value of 109.6 is greater than 7.82
- At 3 degrees of freedom
- The probability of getting chi-square value is between $p < 0.001$
- $P < 0.5$ (or less than 5% probability) which is the cut off probability
- Hence difference between observed and expected results is significant
- and not due to chance

Question 5 [12 marks]

(a) Fig 5.1 below shows the blood glucose concentration measured.

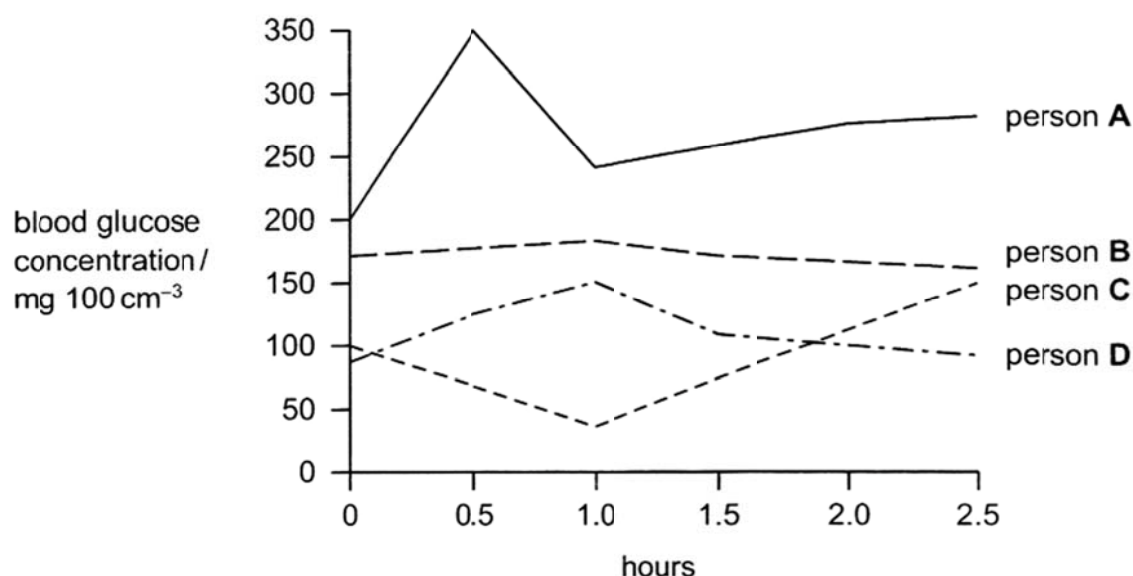


Fig 5.1

(i) Identify which individual has a normal homeostatic system in controlling blood glucose level. Explain your answer. [2]

- Individual D;
- Increase in blood glucose level from 90 mg to 150 mg 100 cm⁻³ of blood from 0 – 1.0 hour as glucose is absorbed into the bloodstream;
- Beta cells of the pancreas detects → produces/triggers insulin production;
- increases glucose permeability in liver and muscle cells;
- return glucose level back to set point 100 mg per 100 ml blood after 2.5 hours

(ii) Describe and explain the condition of person C in the period between 0 – 1.0 hours. [2]

- Hypoglycaemia;
- blood glucose level drop from 100 to 35mg 100 cm⁻³ of blood; (below set point of 90 mg / 100 ml of blood)
- Person C is unable to absorb the glucose from the intestine into his blood stream/due to intensive exercise/fasting; (AVP)
- Glucagon not secreted/synthesised

(iii) State the type of receptor that glucagon binds to and explain how this receptor is the fully activated. [2]

- G-protein linked receptor;
- Binding of glucagon to complementary binding site on receptor;
- cause change in conformation of the receptor which can now bind G-protein

(b) Ras is a GTP-binding protein that helps to broadcast signals from the cell surface to other parts of the cell to stimulate cell proliferation. If Ras function is inhibited,

the cell proliferation responses normally induced by the activated receptors do not occur. Conversely, if a hyperactive mutant, Ras protein, is introduced into some cell lines, the effect on cell proliferation is the same as that induced by the binding of ligands to cell surface receptors. This hyperactive product of mutant ras gene promotes the development of cancer.

Figure 5.2 below shows the sequence of events in a Ras cell cycle-stimulating pathway.

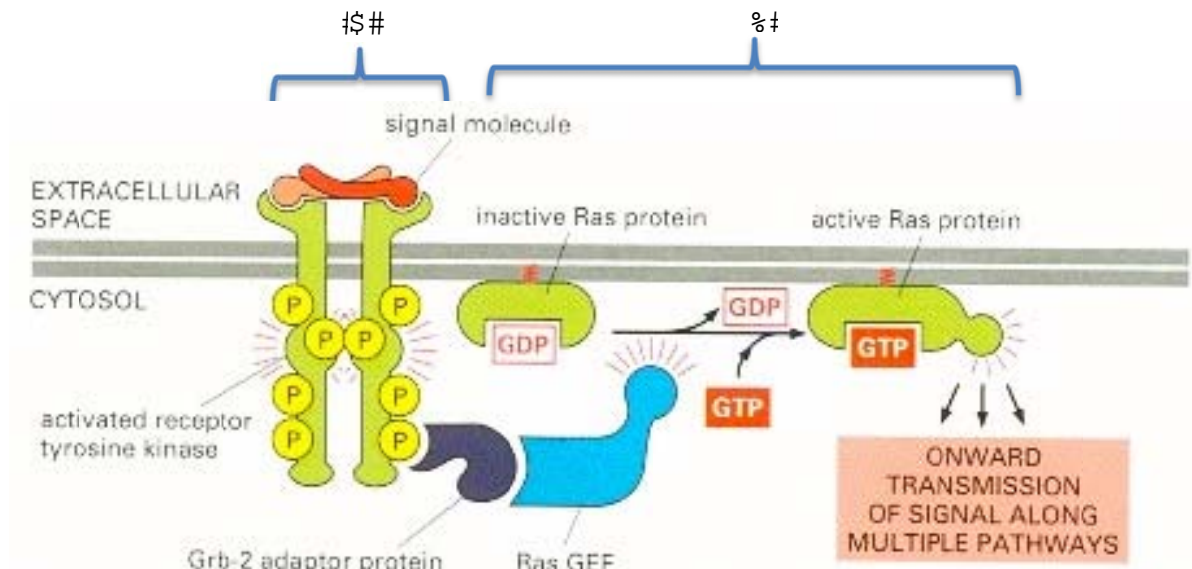


Figure 5.2

- (i) Briefly describe the stages A and B shown in Figure 5.2. [4]

A

- binding of signal molecule to the 2 receptors causes 2 receptor subunits / polypeptide chains / monomers / molecules to form a dimer / dimerise;
- activation of the receptor tyrosine kinase region leads to
- cross phosphorylation of tyrosine amino acids on each other's tyrosine amino acid tails
- and subsequent phosphorylation residues on their respective tail of tyrosine amino acid subunits → activates tyrosine amino acids → making them recognizable as docking sites for relay proteins

B

- relay protein(s), Grb-2 adaptor protein recognizes and binds to specific phosphorylated tyrosine(s) which acts as docking protein
- for Ras GEF which becomes activated
- which activates G protein, Ras protein by removing GDP and
- replacing it with GTP
- in order to activate Ras protein
- to relay signal by activating other proteins or enzymes to bring about various cellular responses (= cell proliferation).

- (ii) Ras protein is a plasma membrane bound G-protein. Suggest how the activated G-protein can relay the signal from the ligand to activate cellular responses. [2]

- a. **By activating enzymes such as adenylyl cyclase which will catalyse the production of second messengers, such as cAMP**
- b. **cAMP will activate protein kinase A resulting in a phosphorylation cascade
→ activate proteins necessary for cellular responses**

(c) Suggest one function of signal transduction. [1]

- a. **may transform the signal into a different form, which is suitable for passing or stimulating a cell response**
OR
- b. **It may amplify the signal it receives either by producing large amounts of a small signal intracellular mediators or by activating many copies of a downstream signaling protein.**

Question 6 [11 marks]

Figure 6.1 shows an electromicrograph of a chloroplast

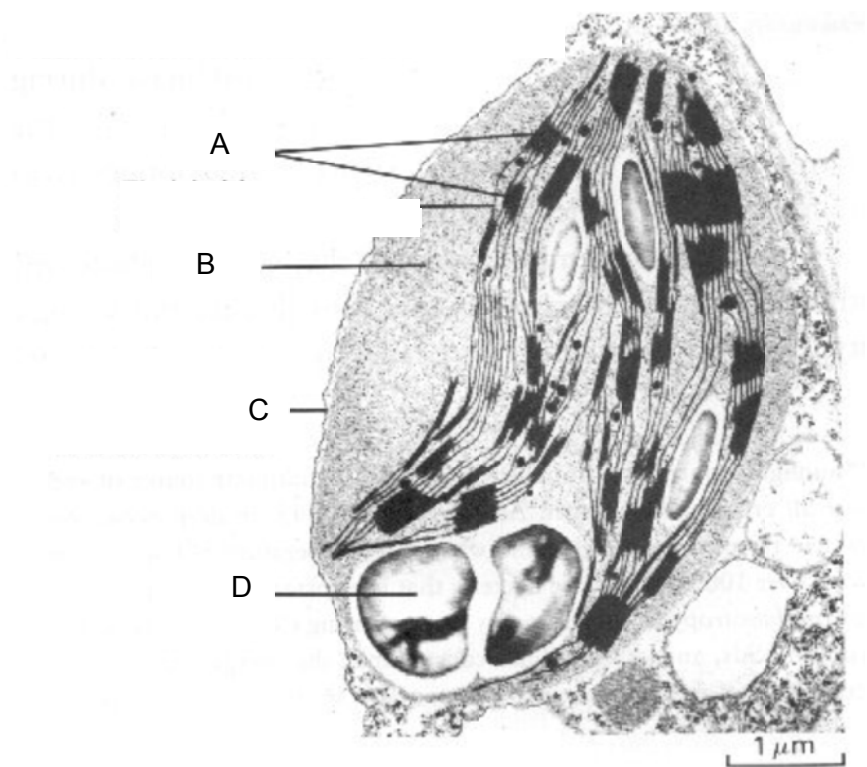


Figure 6.1

(a) Identify the features: [2]

- A grana/thylakoid membranes** **B stroma**
C chloroplast envelope **D starch grain**

(b) The blue dye DCPIP can be converted to colourless reduced DCPIP by gaining electrons.
A suspension of chloroplasts was made by grinding fresh leaves in buffer solution and centrifuging the mixture. Tubes were then prepared and treated in the following way:

Tubes	Content	Treatment	Colour	
			Start	After 20 min
A	1 cm ³ chloroplast suspension 5cm ³ DCPIP	Illuminated strongly	Blue-green	Green
B	1 cm ³ buffer solution 5cm ³ DCPIP	Illuminated strongly	Blue	blue
C	1 cm ³ chloroplast suspension 5cm ³ DCPIP	Left in the dark	Blue-green	Blue-green

- (i) Explain this observation using your knowledge on light-dependent reactions of photosynthesis. [4]
 - a. **DCPIP acts like final hydrogen ion and electron acceptor, NADP^+ in non-cyclic photophosphorylation of the light-dependent reaction;**
 - b. **Tube A: When light and chloroplast suspension are present, photoexcited electrons from water/PSII/PSI are passed through the ETC to DCPIP to reduce it, changing it from blue to colourless.**
 - c. **However, when tube C is placed in the dark, DCPIP remained blue, indicating that it is not being reduced $\rightarrow$ no photoactivation in the dark**
 - d. **DCPIP is thus an indicator of whether photoactivation and thus electron flow through the ETC during light-dependent reaction has taken place.**
 - e. **Tube B shows that DCPIP is not reduced and remains blue;**
 - f. **Proving that chloroplast suspension is required for reduction of DCPIP;**
- (iii) Explain, using tubes A and C, the effects of illumination on Calvin cycle (dark reaction). [2]
 - a. **Light dependent reaction taking place in tube A but not in C means that ATP & NADPH can be produced in A**
 - b. **With ATP and NADPH, Calvin cycle can take place giving rise to sugars**
 - c. **No Calvin cycle in tube C**
 - d. **ATP and NADPH are required for reduction and regeneration phase in Calvin cycle.**
- (iv) The chloroplast suspension may be contaminated with mitochondria. Explain the evidence from the investigation that presence of mitochondria was not responsible for the reduction of DCPIP. [2]
 - a. **Respiration in the mitochondria occurs under both dark and light conditions, producing hydrogen ions and electrons for reduction of NAD^+ ;**
 - b. **When left in the dark/ absence of light/ tube C, DCPIP remained blue-green like at the start of the experiment, indicating that there is no reduction of DCPIP#**

Question 7 [11 marks]

The Galápagos tortoise or Galápagos giant tortoise (*Chelonoidis nigra*) is the largest living species of tortoise and 13th-heaviest living reptile, reaching weights of over 400 kg and lengths of over 1.8 metres. With life spans in the wild of over 100 years, it is one of the longest-lived vertebrates.

The tortoise is native to seven of the Galápagos Islands, a volcanic archipelago about 1,000 km west of the Ecuadorian mainland.

Shell size and shape vary between populations. On islands with humid highlands, the tortoises are larger, with domed shells and short necks. Due to the climate, there is abundant vegetation and vegetation is closer to the ground. On islands with dry lowlands, the tortoises are smaller, with "saddleback" shells, which are flatter than domed shells, and long necks. There is less vegetation. Charles Darwin's observations of these differences on the second voyage of the Beagle in 1835, contributed to the development of his theory of evolution.

Tortoise numbers declined from over 250,000 in the 16th century to a low of around 3,000 in the 1970s. This decline was caused by exploitation of the species for meat and oil, habitat clearance for agriculture, and introduction of non-native animals to the islands, such as rats, goats, and pigs.

(a) (i) Explain how the distinct phenotypic differences between the populations may have arisen. [4]

- a. **Within the population, there is inherited variation/variations in alleles.**
- b. **Selection pressure is the availability of food source on the different islands.**
- c. **Results in differential survival and reproductive rates in different varieties.**
- d. **Islands that have humid highlands have larger tortoises with short necks as vegetation is abundant and closer to the ground hence the tortoises with short necks are more highly adapted to the environment and those with longer necks do not survive as well.**
- e. **Islands that have dry lowlands have smaller tortoises as the food is less abundant and unable to grow to larger sizes.**
- f. **Those with longer necks are better adapted to the environment as they are able to stretch further to reach food that is higher off the ground.**
- g. **Individuals with the advantageous traits will survive better to reproduce and pass on alleles for these traits to the offspring.**
- h. **Islands - Geographical isolation therefore the tortoises are not able to interbreed → no gene flow between populations**
- i. **Genetic drift might have also occurred through founder effect leading to different allele frequencies in the populations on different islands.**

(ii) State the phenomenon that describes the decline in tortoise numbers from the 16th century to the 1970s. [1]

Bottleneck effect

(iii) Suggest a disadvantage to a reduction in population size. [1]

- a. **Decreased genetic variation within the population;**
- b. **Leading to the population being susceptible to new diseases or sudden environmental changes OR**

Increase in the occurrence of recessive diseases as each parent will most probably have a copy of the recessive allele.

Figure 7.1 below shows the relationship between number of substitutions in cytochrome c between related pairs of species and how long ago the two compared species had diverged from a common ancestor.

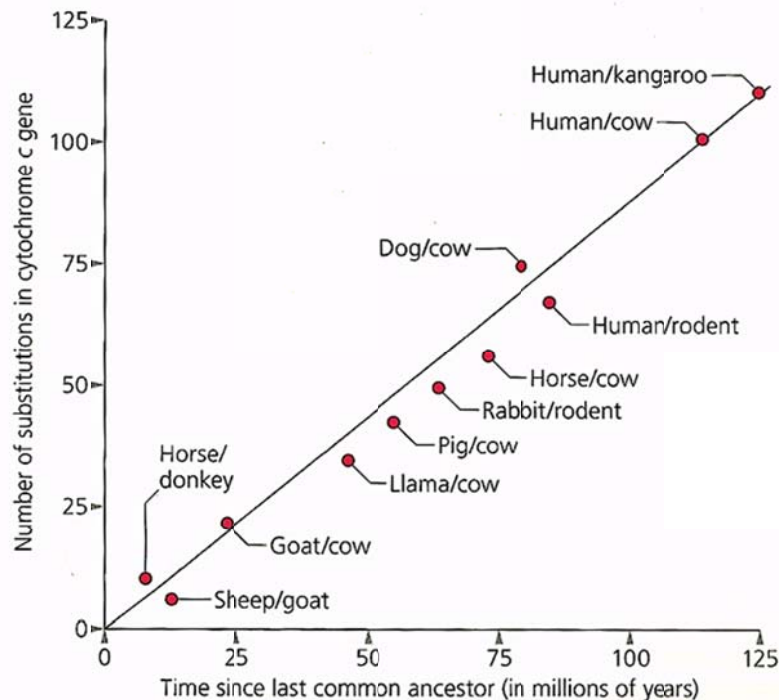


Figure 7.1

(b) Describe and explain the conclusions that can be drawn from Figure 7.1. [3]

- The graph suggests a linear relationship between the number of sequence changes and the time of divergence.
- Such a relationship indicates that the observed rate of neutral mutations remains constant over millions of years.
- The number of nucleotide differences is higher when two species shared a common ancestor in the very distant past than it is in another two species that shared a more recent common ancestor.
- Human and cow have a difference of 90 nucleotides and they diverged from the common ancestor about 120 millions of years ago.
- Compared to horse and donkey which have a difference of 10 nucleotides and they diverged from the common ancestor 8 million years ago. (Any other valid comparison)
- The gene sequences of species which had diverged from each other for a longer period of time would have accumulated more independent mutations, which make their sequences more different.

(c) Suggest the difficulties that arise when using the molecular clock. [2]

- No gene marks time with complete precision. In fact, some portions of the genome appear to have evolved in irregular fits and starts that are not at all clocklike.

- b. Even genes that seem to have reliable molecular clocks are accurate only in the statistical sense of showing a fairly smooth average rate of change. Over time, there may still be chance deviations above and below that average rate.
- c. The same gene may also evolve at different rates in different groups of organisms, making it necessary to calibrate and use molecular clock with care.
- d. Even among genes that are clocklike, the rate of the clock may vary greatly from one gene to another; some genes evolve a million times faster than others.

Section B: Essay

Question 8

(a) Explain how the structure of triglyceride is related to its function. [7]

Chemical nature	Function
Proportionately more H atoms compared to oxygen	- Release more hydrogen ions when oxidized during respiration - Allows production of twice the amount of energy as compared to same unit mass of carbohydrates - Yields twice the amount of metabolic water when oxidized during respiration
More C atoms and H atoms than O atoms → lower molecular weight than eg. carbohydrates	- Lower density/molecular mass → more weight efficient compared to carbohydrates - Favours <u>locomotion of animals</u> - Less dense → Provides <u>buoyancy</u> especially for aquatic animals - Favours <u>seed dispersal</u> since lightweight as it can be stored in large amounts in the seed endosperm.
Hydrophobic / non-polar; thus chemically inert in cells	<u>Long term storage of energy</u> → good for hibernating animals - Can surround and protect vital organs on a long term basis to <u>absorb mechanical shock</u> in case of accidents or falls <u>Good heat insulators</u> <u>Maintenance of body temperature.</u>
Non-polar so will repel ions or charges	- High electrical resistance – <u>good electrical insulators.</u> - Ensures <u>rapid transmission of nerve</u>

	<u>impulses in myelinated nerve cells</u>
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(b) Compare and contrast between collagen and cellulose. [7]

	Collagen	Cellulose
Monomers	Amino acids	Only B-glucose molecules
Bonds between monomers	Peptide bonds formed between amino acid monomers	B 1,4, glycosidic bonds formed between B-glucose monomers
Structure adopted by each chain	Loose helical structure of polypeptide chain due to the lack of intra-chain H bonds (because of lack of N-H group in proline)	Linear chain due to rotation of alternate glucose residue, resulting in bulky side chains on each side of the chain
Subunit making up macromolecule	Tropocollagen – 3 polypeptide chains wound tightly together	Linear unbranched chain
Arrangement of subunit in macromolecule	Tropocollagen arranged in staggered out of step manner to increase tensile and mechanical strength	Long chains run parallel to each other, but not in a staggered manner
Bonds involved in stabilizing macromolecule	Covalent cross-linkages formed between the C-terminus of one tropocollagen and the N-terminus of another tropocollagen	Cumulative strength of H-bonds formed between the hydroxyl groups projected outwards in all directions of each chain
Organism	Produced in animals	Produced in plants
Similarities		
Both serve as structural support with high tensile and mechanical strength		
Both are insoluble in water due to the lack of hydroxyl groups to form H-bonds with water.		

(c) Some enzymes function optimally in acidic conditions, while other enzymes function optimally in alkaline conditions. Explain why different enzymes have different optimal pH. [6]

- Each enzyme is a globular protein
- with a structurally and chemically specific active site.
- This is due to the specific sequence of amino acids in the primary structure
- which determines the types of bonds and location of R groups present on the amino acids in the protein
- The conformation of the active site of each enzyme will change when pH changes
- due to changes to the catalytic, contact and structural residues.
- When the pH increases beyond the optimal pH, positively charged R groups at active sites of enzyme X are neutralised by the abundant OH⁻ in the solution
- As pH decreases to low pH for enzyme Y, the negatively charged R groups at the active sites of enzyme Y are neutralised by the H⁺ in the solution
- Both lead to disruption of ionic bonds and hydrogen bonds
- which in turn disrupt the secondary and 3D conformation (tertiary) & quaternary structures of the enzyme

- k. and thus the conformation of the active site can no longer be maintained.
- l. Substrate no longer fits into active site as enzyme is denatured.

Question 9

- (a) Explain the role of the F plasmids in bacterial conjugation and suggest the advantages of conjugation over transduction. [8]

Explain the role of the F plasmids in bacterial conjugation: (max 4 marks)

- 1 Conjugation - transfer of genetic material (in one direction) through (direct) cell to cell contact between two bacterial cells;
- 2 Ref to one with the F plasmid (donor) and the other without (recipient);
- 3 F plasmid contains genes / codes for synthesis of sex pilus;
- 4 Used for attachment to recipient cell
- 5 And for formation of mating bridge/conjugation tube;
- 6 Ref. to F factor replicates by rolling-circle mechanism (*idea of replication is impt*);
- 7 (single-strand of) F plasmid is transmitted through mating bridge to recipient;
- 8 transferred ssDNA is replicated in the recipient cell, producing a ds F plasmid;
- 9 Conjugation tube breaks;
- 10 (if entire F plasmid was transferred) Recipient cell now has F plasmid/ is F⁺;

Suggest the advantages of conjugation as compared to transduction [max 4]

- 1 No need for 3rd party / virus / vice versa;
- 2 Host cell remains alive;
- 3 Does not rely on chance that wrong DNA fragments are packaged (in generalised transduction);
- 4 Does not rely on chance that viral DNA excises imprecisely (in specialised transduction);
- 5 Homologous recombination not needed unlike (general) transduction (though specialised transduction does not require it);
- 6 Higher chance to transfer genetic material (if cells contact each other) (as transduction relies heavily on chance);
- 7 In transduction, genetic material may or may not be degraded / vice versa;
- 8 Larger portion of DNA/genetic material can be transferred;
- 9 Ref. defined genes (of F plasmid) transmitted during conjugation vs random DNA fragments (via general transduction) OR only genes/DNA fragments beside prophage (via specialised transduction)

- (b) Explain how the lac operon is regulated by both negative and positive control. [8]

- a. Only when lactose is in high supply AND glucose is in low supply does *E coli* use lactose as an energy source.
- b. When lactose is present, some of the lactose gets transported into the cells
- c. Lactose is converted into the inducer, allolactose.
- d. Allolactose is the isomer form of lactose.
- e. The inducer binds to the repressor protein and alters its conformation
- f. so it cannot bind to the operator site (repressor becomes inactive)
- g. In the absence of an operator-bound repressor protein, RNA polymerase can now transcribe the structural genes into mRNA,
- h. which are then translated into enzymes, permease, transacetylase and β -galactosidase, which transport and hydrolyse the disaccharide, lactose.
- i. The *lac* operon has a very inefficient promoter element – that is, it has a low

- affinity for RNA polymerase, even when the repressor protein is inactivated.
- j. cyclic AMP (cAMP), accumulates when glucose is in low concentration in the cell.
 - k. Another regulatory protein, catabolite activator protein (CAP) (which is produced in inactive form) can bind to CAP-binding site.
 - l. To activate CAP, cAMP must bind to CAP;
 - m. The resulting cAMP-CAP complex then binds to the CAP-binding site.
 - n. This binding of active CAP bends DNA's double helix strengthens the affinity of the promoter region for RNA polymerase
 - o. the rate of transcriptional initiation increases in the presence of lactose.
 - p. Lac operon is thus regulated by both negative (due to repressor) and positive (due to activator) control.

(c) Suggest why operons are necessary in bacteria. [4]

- a. bacteria genome is short;
- b. having operons saves space;
- c. as a few genes share one promoter;
- d. producing polycistronic mRNA;
- e. allows transcription of a few related genes at the same time (better regulation of gene expression);
- f. being single cellular also means that the bacteria is easily affected by the fluctuating environment;
- g. allows bacteria to adapt to environment quickly as the related genes are transcribed simultaneously;
- h. better use of resources, thus leads to less wastage of energy and resources